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Course: Data Science
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Exercise: 2.1
Topic: ~~SQL Date Functions~~ Advanced SQL
NULL Functions

Table 1: Employees

employee-id	name	department	salary
1	Alice	HR	5000
2	Bob	IT	NULL
3	Charlie	NULL	7000
4	Dana	Finance	NULL

1. IF NULL, display 0

```
SELECT employee-id, name,
       IF NULL (salary, '0') AS salary-with-default
FROM employees;
```

employee-id	name	salary-with-default
1	Alice	5000
2	Bob	0
3	Charlie	7000
4	Dana	0

2. IF NULL, show 'Not Assigned'.

```
SELECT employee-id, name,
       IF NULL (department, 'Not Assigned') AS department_name
FROM employees;
```

employee-id	name	department_name
1	Alice	HR
2	Bob	IT
3	Charlie	'Not Assigned'
4	Dana	Finance

Table 2 : Orders

order_id	customer_id	delivery_date
101	201	2024-12-01
102	202	NULL
103	NULL	2024-12-03

3. Orders with NULL customer_id (IS NULL ())

```
SELECT order_id, customer_id
FROM orders
WHERE customer_id IS NULL;
```

order_id	customer_id
103	NULL

4. Show all orders, If delivery_date is NULL, show 'Pending'.

```
SELECT order_id, customer_id
IF NULL (delivery_date, 'Pending') AS delivery_status
FROM orders;
```

order_id	customer_id	delivery_status
101	201	2024-12-01
102	202	Pending
103	NULL	2024-12-03

Table 3: Students

student_id	name	grade
1	Ethan	85
2	Maya	NULL
3	Olivia	90

5. Show all students and grades. Replace NULL with '0'.

```
SELECT student_id, name
       IFNULL (grade, '0') AS final_grade
FROM   students;
```

student_id	name	final_grade
1	Ethan	85
2	Maya	0
3	Olivia	90

6. Students who haven't been graded

```
SELECT * SELECT COUNT (*), final_grade AS not_graded_count
FROM   students
WHERE  final_grade IS NULL;
```

not_graded_count
1

Table 4: Products

product_id	name	price	discount
S01	keyboard	25	NULL
S02	mouse	15	5
S03	monitor	100	NULL

7. Show product name, price, and final price after discount (assume 0 if discount is NULL).

~~SELECT product_id, name, final_price
FROM products~~

SELECT product_id, name
IFNULL (discount, 0) AS final_price
FROM products;

product_id	name	final_price
S01	keyboard	25
S02	mouse	14,25
S03	monitor	100

Table 5: Customers

customer_id	name	email
1	Linda	NULL
2	Joseph	joseph@mail.com
3	Nia	NULL

SELECT COUNT(email) AS missing_email_count
FROM customers
WHERE email IS NULL;

missing_email_count
2

9. Show customers with email, IF NULL display 'No email'

```
SELECT customer_id, name,
       IFNULL(email, 'No email') AS email_display
FROM customers;
```

customer_id	name	email_display
1	Linda	No email
2	Joseph	joseph@mail.com
3	Mia	No email

Table 6: Payments

payment_id	method	status
301	Credit	NULL
302	PayPal	Success
303	NULL	Failed

10. Show payment details with method replaced by "Unknown" if NULL

```
SELECT payment_id, status,
       IFNULL(method, 'Unknown') AS method_display
FROM payments;
```

payment_id	method_display	status
301	Credit	NULL
302	PayPal	Success
303	Unknown	Failed

11. Table 7: Inventory

```
SELECT item_id, item_name,  
       IFNULL(quantity, '0') AS quantity_checked  
FROM   inventory;
```

item_id	item_name	quantity_checked
1	Pen	0
2	Notebook	150
3	Eraser	0

12. Table 8: Employees_Extra

```
SELECT emp_id,  
       COALESCE(bonus, commission, 0000) AS first_available_reward  
FROM   Employees_Extra;
```

emp_id	first_available_reward
1	300
2	100
3	0000

13. Table 9: Classes

```
SELECT COUNT(room) AS no_room_count  
FROM   classes  
WHERE  room IS NULL;
```

no_room_count
2

14. Table 10: Attendance

```
SELECT student_id, date,
       IF NULL (status, 'not marked') AS attendance_status
FROM attendance;
```

student_id	date	attendance_status
1	2025-04-01	not marked
2	2025-04-01	present
3	2025-04-01	absent

15. Table 11: Bank_accounts

```
SELECT account_id,
       IF NULL (account_type, 'Unknown') AS type_display,
       AND IF NULL (balance, '0') AS balance_checked
FROM Bank_accounts;
```

account_id	type_display	balance_checked
A1	Savings	0
A2	Current	5000
A3	Unknown	2000

16. Table 12: Projects

```
SELECT project_id, title,
       IF NULL (start_date, 'TBD') AS start_display
FROM projects;
```

project_id	title	start_display
1	Website Revamp	2025-01-10
2	Mobile App	TBD
3	Data Migration	TBD

17. Table 13 : Reviews

```
SELECT review_id, product_id,
       COALESCE (comment, 'No comment') AS comment_display,
       COALESCE (rating, 0) AS rating_display
FROM reviews;
```

review-id	product-id	comment-display	rating-display
1	S01	Great product	4
2	S02	No comment	0
3	S03	Works fine	3

18. Table 14 : Suppliers

```
SELECT supplier_id, name,
       COALESCE (phone, alt_phone, 'no contact') AS contact_number
FROM suppliers;
```

supplier-id	name	contact-number
1	Global Goods	123 456 789
2	Best Supplies	987654321
3	ValueSource	no contact

19. Table 15 : User-Settings

```
SELECT user_id,
       IF NULL (theme, 'light') AS theme_set,
       IF NULL (language, 'English') AS language_set,
       IF NULL (timezone, 'UTC') AS timezone_set
FROM User-Settings;
```


user_id	theme_set	language_set	timezone_set
1	light	English	UTC
2	dark	English	UTC+1
3	light	English	UTC

20. Table 16: Maintenance

```

SELECT record_id, machine_id,
       IFNULL(issue, 'Unknown Issue') AS issue_log,
       IFNULL(technician, 'Not assigned') AS technician_name
FROM Maintenance;

```

record_id	machine_id	issue_log	technician_name
1	M101	Overheating	not assigned
2	M102	Unknown-issue	Not assigned
3	M103	Jammed	Alex